7.3.2 Use protractor, set squares and ruler to construct square and rectangle with given sides.

Example 1: With the help of ruler and protractor construct a rectangle of sides 4.5 cm and 2.5 cm.

Steps of construction:

Step I: Use ruler, draw $\overline{EF}$ of 4.5 cm.

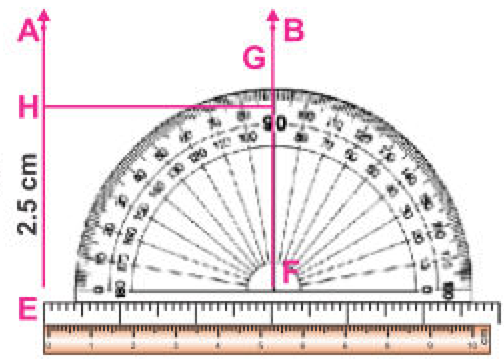
Step II: At point F, draw $\angle EFB = 90^\circ$ with the help of protractor.

Step III: Using ruler cut $\overrightarrow{FB}$ such that $\overline{FG}$ of 2.5 cm

Step IV: At point E, draw $\angle FEA = 90^\circ$ with the help of protractor.

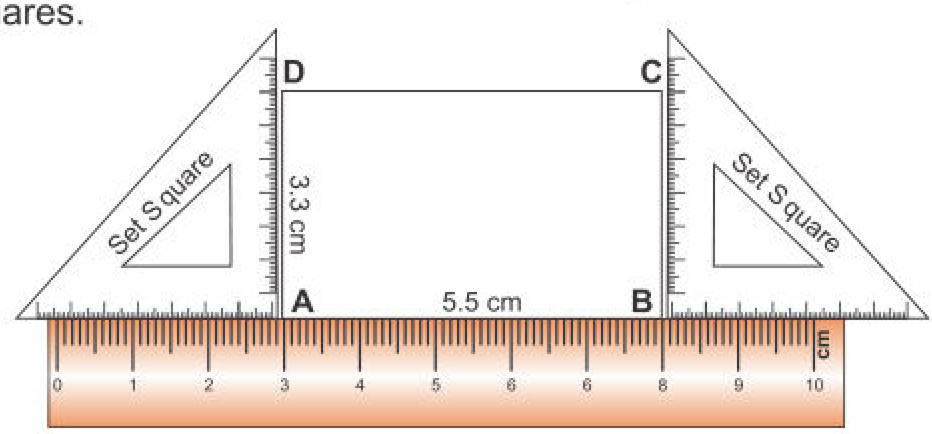
Step V Using ruler, cut $\overrightarrow{EA}$ such that $\overline{EH}$ of 2.5 cm.

Step VI: Draw $\overline{GH}$ by joining the points G and H i.e. draw $\overline{GH}$ of 4.5 cm.



Hence quadrilateral EFGH is the required rectangle.

Example 2: To construct a rectangle of which the length is 5.5 cm and breadth is 3.3 cm with the help of ruler and a set squares.



Teacher's Note Teacher should help the students how to use protractor, set square and ruler for construction of rectangle.

Steps of construction:

Step I: Use ruler, draw $\overline{AB}$ of 5.5 cm length.

Step II: Place both set squares on $\overline{AB}$, such that their right angled perpendicular sides are parallel to each other.

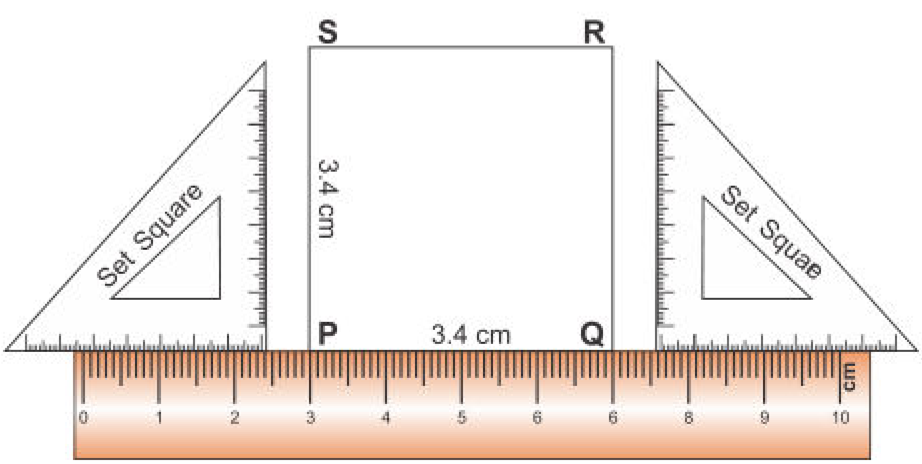
Step III: With the help of set squares, construct a perpendicular at point A. Cut it 3.3 cm and mark it point D.

Step IV: Construct perpendicular with the help of set squares at point B. Draw another perpendicular $\overline{BC}$ and cut it 3.3 cm length and mark it point C.

Step V Join point C and D with the help of ruler and draw $\overline{CD}$.

Thus quadrilateral ABCD is the required rectangle.

Example 3: Construct a square of which the length of a side is 3.4 cm with the help of ruler and a set square.



Steps of construction:

Step I: Use ruler, draw $\overline{PQ}$ = 3.4 cm long.

Step II: Place both set squares on $\overline{PQ}$, such that their right angled perpendicular sides are parallel to each other.

Step III: With the help of set square, draw a perpendicular on point P. Measure it 3.4 cm and mark it as point S.

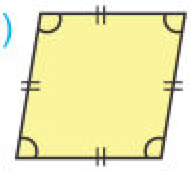
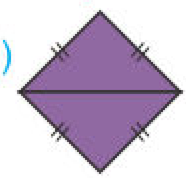
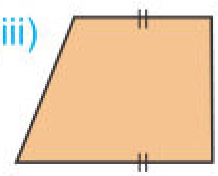
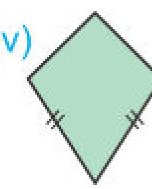
Step IV: Construct another perpendicular at point Q. Cut it 3.4 cm and mark it point R.

Step V: Use ruler, Join point R and S then draw $\overline{RS}$ = 3.4 cm long.

Thus quadrilateral PQRS is the required square.

EXERCISE 7.7

1. Recognize the names and give the answers of the following for each quadrilateral.

(i)  (ii)  (iii)  (iv) 

__________ __________ __________ __________

How many?

(i) Sides ________ names ________________________

(ii) Angles ________ names _______________________

(iii) Vertex _________

(iv) Measurement of each angle__________________

2. Construct rectangle with sides given below with the help of a ruler and a protractor.

(i) 4 cm, 3c m (ii) 6 cm, 3.5 cm (iii) 5.5 cm, 2.8 cm

3. Construct a square with a side given below with the help of a ruler and a protractor.

(i) 3 cm (ii) 4 cm (iii) 5.4 cm

4. Construct rectangles with sides given below with the help of a ruler and set squares.

(i) 5 cm, 4 cm (ii) 6 cm, 3 cm (iii) 4.6 cm, 3.5 cm

5. Construct a square with a side given below with the help of a ruler and a set squares.

(i) 4 cm (ii) 5 cm (iii) 4.4 cm

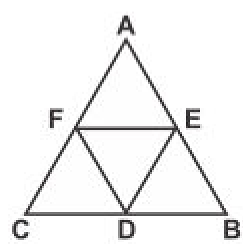
REVIEW EXERCISE 7

1. Marium and Sakina start from a point A. Marium moves towards east up to E and Sakina moves towards south up to S. Draw their paths and name the kind of angle which will be formed between them.

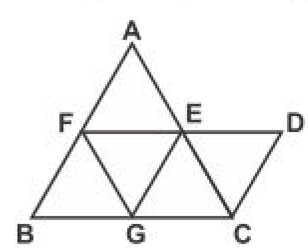
2. State the kinds of angle that is formed between the following directions:
   (i) East and West (ii) East and North
   (iii) From North to West through East

3. Is it possible to have a triangle, in which:
   (i) Two of the angles are right angels?
   (ii) Two of the angles are acute?
   (iii) Two of the angles are obtuse?
   (iv) Each angle is less than 60°.

4. How many quadrilaterals are formed in the adjoining figure? Name them.



5. Write the names of the triangles and quadrilaterals formed in the figure. Also mention their kinds.



6. Prove that a square is a rectangle but a rectangle is nota square. Give reasons in your own words.

7. Two angles of a triangle are of measure 65°and 45.°Find the measure of the third angle.

Unit 8

PERIMETER AND AREA

8.1 PERIMETER AND AREA

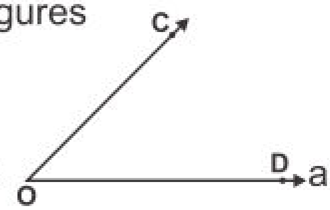
8.1.1 Recognize region of a closed figure.

There are two types of figures in geometry.

(a) Open figures (b) Closed figures

(a) Open figures:

A line 

and an angle 

are examples of open figures.

Here are some more examples of open figures.

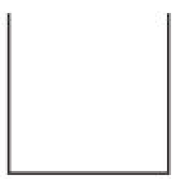


Fig. 1

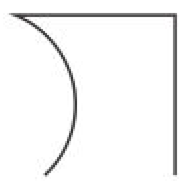


Fig. 2

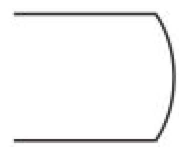


Fig. 3

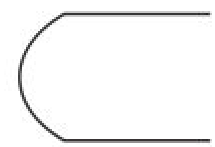


Fig. 4

We can not determine the region enclosed by the open figures because it is open on at least one side.

(b) Closed figures

Consider an other figure (i). It is closed figure represented by the triangle XYZ. It shows the triangular region which is marked by dots. $\overline{XY}$, $\overline{YZ}$, and $\overline{ZX}$ form the boundary of the triangular region XYZ. The sides $\overline{XY}$, $\overline{YZ}$, and $\overline{ZX}$ are the parts of triangular region.

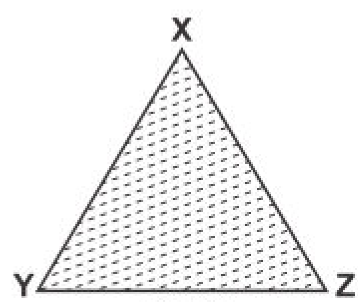


Fig. (i)

We can also show the closed figure in the form of the circular region figure (ii). The boundary of this circular region is the circle itself.

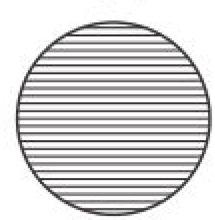
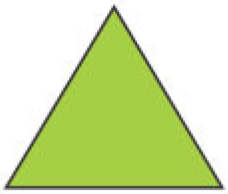


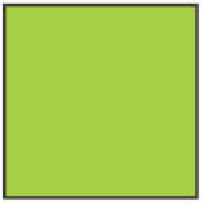
Fig. (ii)

Teacher's Note Teacher should demonstrate the open and closed geometrical figures through thread/rope.

8.1.2 Differentiate between perimeter and area of a region.

Look at the following figures.


A triangular region


A square region

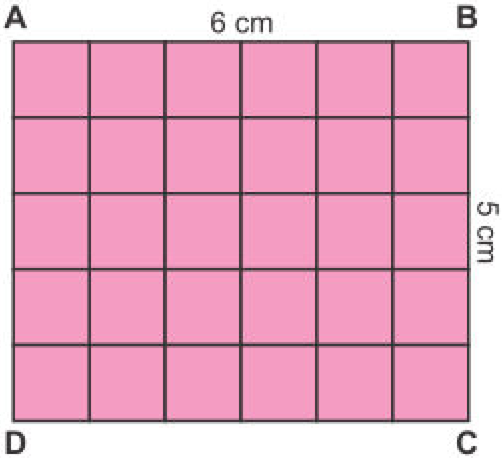

A rectangular region

Here we see that these regions are bounded of line segments. This means that we find the distance around the figure or the length of the boundary which is known as the **perimeter** of the figure.

"Perimeter" is the measure of the total length of the sides, or of any closed figure. It is the length of the sides, or the boundary of the region.

"Area" is the measure of the region within a closed figure. We can find the area of a closed figure by fitting unit squares in the region.

Example: Find the perimeter and area of rectangle with sides of 5cm and 6cm.



Solution:

Length = 6 cm

Breadth = 5 cm

Perimeter = B + L + B + L

= 5 + 6 + 5 + 6

= 22 cm

In given rectangle there are total 30 unit squares. So, the area of ABCD = 30 squares cm.

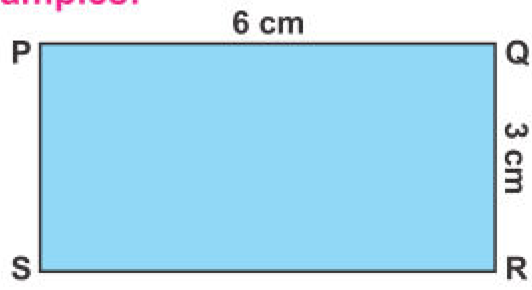
8.1.3 Identify units for measurement of perimeter and area.

(a) Units for the perimeter

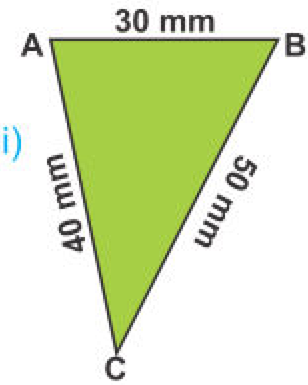
Look at the following figures:

Examples:

(i)



(ii)



Here PQRS is a rectangle.

$m\overline{PQ}$	=	6 cm
$m\overline{QR}$	=	3 cm
$m\overline{SR}$	=	6 cm
$m\overline{PS}$	=	+ 3 cm
Perimeter		18 cm

Here ABC is a triangle.
Remember 10 mm = 1 cm

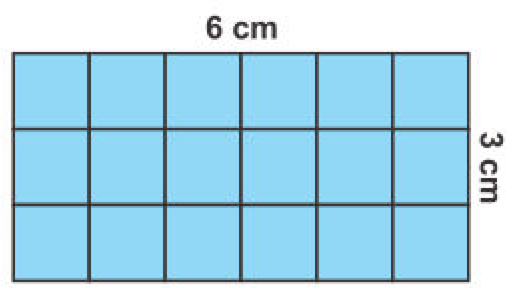
$m\overline{AB}$	=	**30 mm**	=	3 cm
$m\overline{BC}$	=	**50 mm**	=	5 cm
$m\overline{AC}$	=	**40 mm**	=	+ 4 cm
Perimeter				12 cm

= (30 + 50 + 40) mm
= 120 mm = 12 cm

> The unit for the perimeter is the same as the length used, i.e we can use mm, cm, m and km as the units for measurement of the perimeter.

(b) Units of the area

In the figure the length of rectangle is 6 cm and its breadth is 3 cm. We can find its area by fitting each unit squares with one side of 1 cm. There are total 18 unit squares in the rectangle.

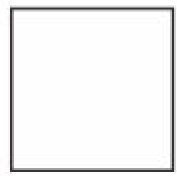


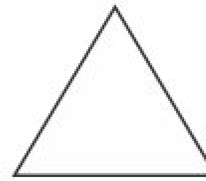
So its area is 18 squares with side of 1 cm each.
i.e Area = 18 sq. cm
Here the unit of area is sq. cm
Similarly sq. m, sq. km, sq. mm are also the units of area.

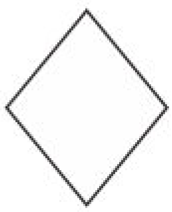
EXERCISE 8.1

A. Look at the following figures carefully. (✓) the shapes which are closed figures and (✗) the open figures.

(1) 

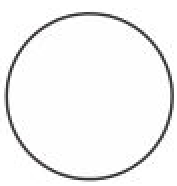
(2) 

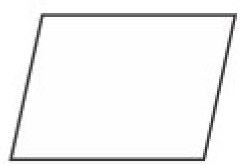
(3) 

(4) 

(5) 

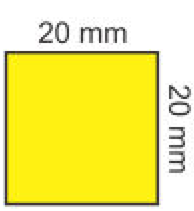
(6) 

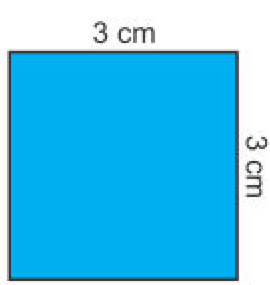
(7) 

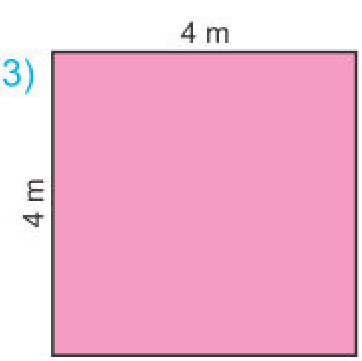
(8) 

(9) 

B. Find the perimeter and write the unit of perimeter for each of the following:

(1) 

(2) 

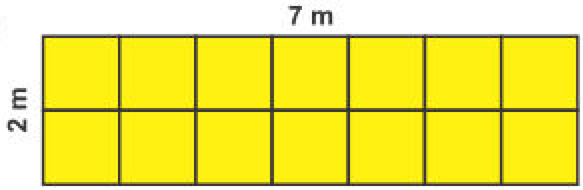
(3) 

(a) Here perimeter ______. (a) Here perimeter ______. (a) Here perimeter ______.

(b) Unit of perimeter _____. (b) Unit of perimeter _____. (b) Unit of perimeter _____.

C. Find the area by counting small unit squares and write the unit of area for each of the following:

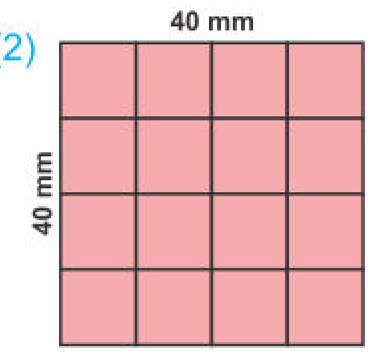
(1)



(a) Area ________ squares.

(b) Unit of area ________.

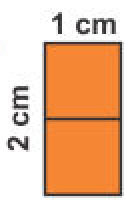
(2)



(a) Area ________ squares.

(b) Unit of area ________.

(3)



(a) Area ________ squares.

(b) Unit of area ________.

(4)

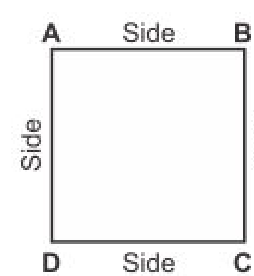
(a) Area _______ squares.

(b) Unit of area ________.

Write and apply the formulas for the perimeter and area of a square and rectangle

(a) Formulas for perimeter of a square and rectangle

(i) We know that a square is a quadrilateral shape in which all its four sides are equal.

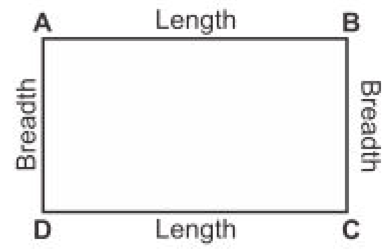


Perimeter of the square

$$\begin{aligned} ABCD &= m\overline{AB} + m\overline{BC} + m\overline{CD} + m\overline{DA} \\ &= \text{side} + \text{side} + \text{side} + \text{side} \\ &= 4 \times \text{side} \\ &= 4 \times (\text{length of a side of square}) \end{aligned}$$

Thus, formula for perimeter of a square figure = 4 x length of side

(ii) Draw a rectangle ABCD. Measure its sides.



Then find its perimeter as follows:

Perimeter of the rectangle ABCD

$= m\overline{AB} + m\overline{BC} + m\overline{CD} + m\overline{DA}$
= length + breadth + length + breadth
= 2 x length + 2 x breadth
= 2 x (length + breadth)

Thus, formula for perimeter of a rectangle
= 2 x (length + breadth)

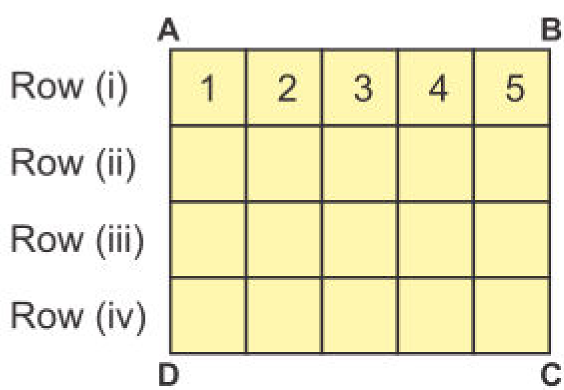
(b) Formulas for area of a square and rectangle

Activity 1

Look at the rectangle ABCD with the length of 5 cm and breadth 4 cm.

Length ________

Breadth ________



(i) What is the length of rectangle ABCD?
(ii) What is the breadth of rectangle ABCD?
(iii) How many rows in the figure?
(iv) How many unit squares in one row?
(v) How many unit squares altogether?

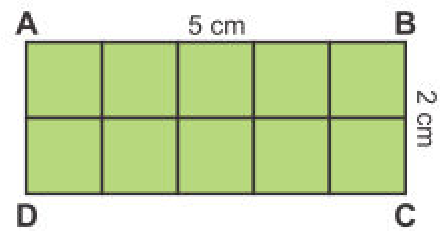
Thus, the area of given rectangle ABCD = 20 cm^2.

So, area of rectangle = Length x Breadth

Thus the formula for finding the area A of a rectangle is:

> Area of a rectangle = Length x Breadth
> or A = L X B

Example. Find area of a rectangle whose length is 5 cm and breadth is 2 cm.



Solution:

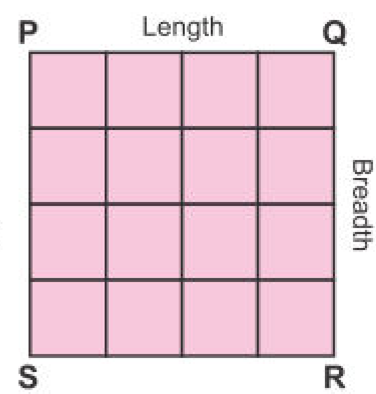
Here length = 5 cm

breadth = 2 cm

Area of a rectangle ABCD

= Length x Breadth.

= 5 x 2 = 10 sq. cm



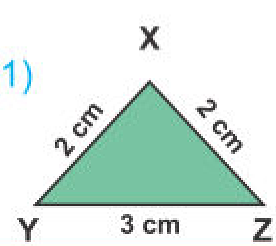
Using formula, find area of the given square shape.

Since the length and breadth of a square are equal, the area of a square PQRS = Side x Side

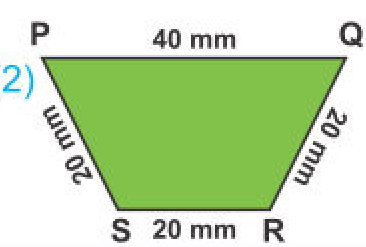
EXERCISE 8.2

A. Find the perimeter of each of the following figures:

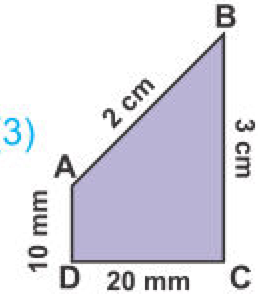
(1)



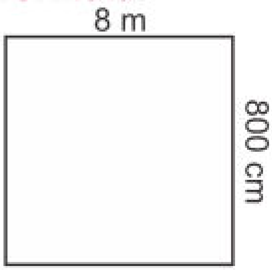
(2)

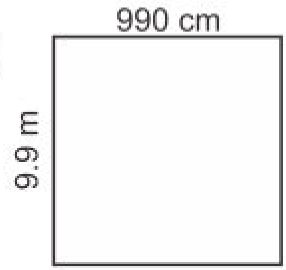


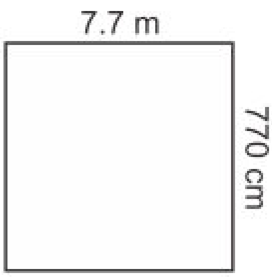
(3)



B. Find the area of the following shapes by using formula.

(1) 

(2) 

(3) 

(4) 

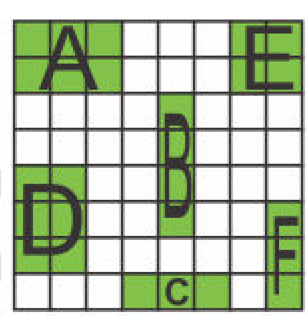
C. Find the area and perimeter of each of the following rectangles.

(1) L = 3 cm, B = 2 cm
(2) L = 5 cm, B = 1 cm
(3) L = 4 cm, B = 3 cm
(4) L = 8 cm, B = 2 cm
(5) L = 9 cm, B = 5 cm
(6) L = 7 cm, B = 4 cm
(7) L = 4.5 cm, B = 2 cm
(8) L = 8 cm, B = 3.5 cm

D. Find area and perimeter of a square whose sides are given below:

(1) 4 cm
(2) 6 cm
(3) 7.5 cm
(4) 8.2 cm
(5) 5 cm 6 mm
(6) 9 cm 2 mm

E. Answer the questions.

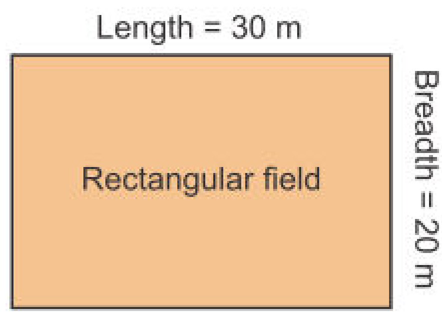


(1) Which shape has the same area as shape B?
(2) Which shape has the same area as shape C?
(3) Which rectangle shapes are equal in area as rectangle A?
(4) Which rectangle shapes are equal in area as square E?
(5) Which shapes have the same area?

Solve appropriate problems of perimeter and area.

Example 1. Length of a rectangular field is 30 m and its breadth is 20 m. Find the perimeter of the rectangular field.

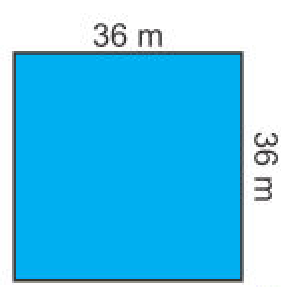
Solution:



Length of the given rectangular field = 30 m
Breadth of the given rectangular field = 20 m
Formula: Perimeter of the given rectangular field is
2 x (Length + Breadth)
= 2 x (30 m + 20 m)
= 2 x 50 m = 100 m

Example 2. A square ground has each side 36 m. What distance will a boy cover by cycling around the ground three times?

Solution: The side of the square ground is 36 m each. Therefore perimeter of the ground is:



4 x side = 4 x 36 m
= 144 m

$$\begin{array}{r} 144 \\ \times\ 3 \\ \hline 432 \\ \hline \end{array}$$

In one time boy will cover 144 m distance.
In three times the boy will cover 144 x 3 = 432 m

Example 3. The perimeter of a rectangular park is 320m. Its length is 70m. Find the breadth.

Solution:

Perimeter = 320m

Length = 70m

Perimeter = 2 x (length + breadth)

Breadth = $\frac{\text{Perimeter}}{2} - \text{Length}$

$= \left(\frac{\cancel{320}^{160}}{\cancel{2}_{1}}\right) - 70 = \frac{160}{1} - 70$

$= 160 - 70 = 90\text{m}$

Example 4. The perimeter of a square shape is 280 cm. Find the length of each side.

Solution:

Perimeter = 280 cm

Length of each side = $\frac{\text{Perimeter}}{4}$

$= \frac{\cancel{280}^{70}}{\cancel{4}_{1}} = \frac{70}{1} = 70\text{ cm}$

Example 5. A rectangular field is 80m long and 60m wide. Find the cost of laying green grass in it; at a rate of Rs 2.50 per sq. Metre. (Remember wide = width = breadth).

Solution:

Rectangular field has length = 80m, width = 60m

Area of rectangular field = Length x Width

= 80m x 60m

= 4800 sq. m

Now let us find the cost of laying green grass in the field.

One sq. metre costs Rs 2.50

4800 sq. metres will cost Rs (4800 x 2.50)

= Rs 12000

EXERCISE 8.3

1. A rectangular park is 84 m long and 56 m broad. Find the perimeter of the park.

2. A square room is 7 m wide. Find the area of the room.

3. A square picture is 60 cm wide. How long wooden frame will be required to reap it form all sides?

4. The length and breadth of a rectangular agriculture field are 190 m and 160 m respectively. Find the area and the perimeter of the field.

5. How much lace is required to fix around a rectangular bed sheet, whose length is 2 m 80 cm and width is 1 m 50 cm?

6. Find the area of both fields; a square of 25 m side and a rectangle 30 m in length and 20 m in breadth.

7. A rectangular agriculture park is 75 m long and 40 m wide. Find the cost of laying green grass in it at the rate of Rs 25 per square metre.

8. Consider a room 15m long and 12m wide. A square carpet 10m x 10m is placed on it.

   (i) What is the area of the floor of the room?

   (ii) What is the area of the carpet?

   (iii) Which is the more in area; the floor or the carpet? And by how much?

REVIEW EXERCISE 8

A. Choose and tick (✓) the correct answer.

1. The space occupied by the boundary of a shape is called:
   (a) Triangle (b) Square (c) Perimeter (d) Region

2. The distance along all the sides of a closed shape is called:
   (a) Triangle (b) Square (c) Perimeter (d) Region

3. The perimetre of a square of length of each side 4 cm is:
   (a) 16 m (b) 16 m^2 (c) 16 cm (d) 16 cm^2

4. The area of a square having each side of 3 cm is:
   (a) 6 cm (b) 9 cm (c) 9 cm^2 (d) 12 cm

5. The area of a rectangle with length 4 cm and breadth 2 cm is:
   (a) 4 cm (b) 8 cm (c) 8 cm^2 (d) 12 cm

6. The perimeter of a rectangle with length 6 cm and breadth 3 cm is:
   (a) 6 cm (b) 18 cm (c) 9 cm (d) 15 cm

B. Answer the following questions.

1. Write formula of area of a square.
2. Write formula of perimeter of a rectangle.
3. Find the perimeter and area of a square of side 7 cm.
4. Find the perimeter and area of a rectangle with length and breadth 8 cm and 5 cm respectively.